



Robots have taken over

2018 seemed to be the year where the robots have taken over and automated several service jobs in major sectors. <https://digit.in/jan19-85>



Sound-proof kennel?

Ford Motot Co. has developed a quiet kennel that blocks loud sounds using noise cancelling technology. <https://digit.in/jan19-86>

ing format due to the optimal tradeoff that it offered between film cost and image quality. Another reason could be the availability of area around the film. It was modified over the years to include digital sound data into almost all of its non-film area, along with modifications for colour capture and multiple widescreen formats.

16MM

Alongside 8mm and 35mm, 16mm is another widely used film format from the past. Generally used for industrial and educational filmmaking, this was also popular for home video and amateur filmmaking alongside 8mm and later, Super 8. Due to the expense of the 35mm format, the 16mm was introduced for the amateurs and was looked



down upon by professionals. It gradually gained capabilities like better sound reproduction and garnered popularity.

16mm has been in popular use until more recently as compared to 8mm. Some of the notable TV series and movies to be shot in it are The Hurt Locker, Moonrise Kingdom, Buffy the Vampire Slayer (first two seasons) and more. In fact, today a number of digital cameras emulate the look and feel of a 16mm film by using 16mm lenses and 16mm-sized sensors. Not being as wide as many other popular formats, 16mm lends itself well to close up shots, but is still better than 8mm for wide shots.

Popular 16mm cameras include Aaton Xtera, Aaton XTRprod, Arriflex 16SR3, and Arriflex 416. Amateurs and

hobbyists are more likely to go for these older models often available as used or second-hand products.

70MM

While some attribute the emergence of 70mm to the film industry's desperate need to innovate in the face of the television gaining popularity, it's not easy to discount the allure of the widescreen. While initial formats like the Cinemarama failed due to being unwieldy, they also showed that the market needed a widescreen technology that was (comparatively) cost effective.

The founder of Cinerama, Mike Todd developed "Todd-AO" which provided better 70mm performance and also allowed for superior sound quality. This allowed six sound channels, later incorporated by Dolby, as the now well-known '5.1' surround sound. Interestingly, 70mm in the years after that saw conversions both ways - 70mm films were often re-released in 35mm format for wider release and highly successful 35mm often got the 'blow-up' treatment for audiences to enjoy them in the 70mm format.

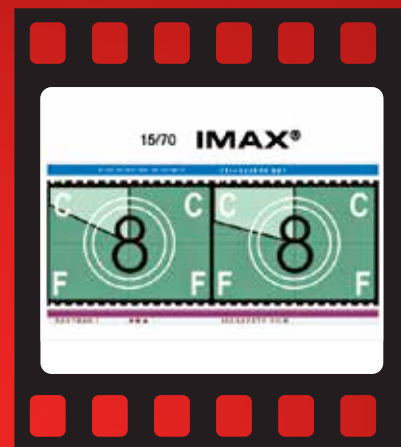
Owing to the high cost and the growing popularity of multiplexes with smaller screens, the 70mm format saw a considerable decline since the 70s. However, a lot of significant modern day advocates of 70mm still stick to the format. Directors like Christopher Nolan, Quentin Tarantino, Kenneth Branagh have used the format in quite a few of their recent movies. A particular benefit of using 70mm over 35mm is



for special effects, where the larger film does a better job at avoiding film grain during optical compositing. However, digital alternatives have taken over that benefit as well.

IMAX

Technically, IMAX is much more than a film format - it is a whole set of cameras, film formats and projectors. However, we're going to focus on the film format here. To put things into perspective, as compared to typical 35mm film, IMAX film has almost three times the horizontal resolution. To achieve this without breaking the capture process,



IMAX film actually passes horizontally through the camera instead of the usual vertical method. It also needs to pass through the camera that much faster, at 102.7 m/min compared to 34m/min of 65mm films. As a result, it requires three times the film stock as compared to a 65mm film of comparable scope.

While initially this meant that IMAX technology would be fairly limiting and difficult to use, in recent years that has changed. In 2011, IMAX announced a 4K 3D camera that would be comparable in resolution to standard IMAX film cameras. Consequently in 2015, they also announced a 4K 2D camera developed with Arri, based on the latter's Arri Alexa 65 technology. While the first movie to use this camera was Captain America: Civil War, Avengers: Infinity War was the first movie to be entirely shot on IMAX. The upcoming Avengers: Endgame will receive the same treatment. 📽️